

PREFIX TUNING FOR AUTOMATED AUDIO CAPTIONING

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ABSTRACT

Audio captioning aims to generate text descriptions from environmental sounds. One challenge of audio captioning is the difficulty of the generalization due to the lack of audio-text paired training data. In this work, we propose a simple yet effective method of dealing with small-scaled datasets by leveraging a pre-trained language model. We keep the language model frozen to maintain the expressivity for text generation, and we only learn to extract global and temporal features from the input audio. To bridge a modality gap between the audio features and the language model, we employ mapping networks that translate audio features to the continuous vectors the language model can understand, called prefixes. We evaluate our proposed method on the Clotho and AudioCaps dataset and show our method outperforms prior arts in diverse experimental settings.

Index Terms— Automated audio captioning, audio representation learning, multi-modal learning, prefix tuning

1. INTRODUCTION

Automated audio captioning (AAC) is the task of describing the input audio with natural language. Compared to other speech-related tasks [1, 2], AAC is characterized by mainly focusing on environmental sounds around us. Text descriptions generated from such sounds provide unrestricted and rich contextual information, which can be further extended to diverse applications, such as interpretable monitoring systems and text-based audio retrieval systems [3].

Following the primary work [4], the typical architecture design for this task is an encoder-decoder. The encoder extracts audio features from the input audio, and the decoder generates a caption conditioned on audio features. Specifically, pre-trained audio models, such as PANN [5] or VGGish [6], are used for the encoder, and the decoder is designed by shallow Transformers [7] or recurrent neural network (RNN) [8]. Diverse combinations of encoder-decoder architectures are proposed [9–14], and additional sub-tasks, such as keyword prediction [15], are employed to improve the performance.

Despite all these efforts, we found that the generalizability issue is less considered and is yet to be resolved. We observe that the prior arts¹ show a significant performance drop when testing on the out-of-domain dataset, *e.g.*, training with one dataset and testing on another. We assume that such degradation is caused by the fundamental lack of training data in AAC. The common datasets in AAC, AudioCaps [16] and Clotho [17], contain 38, 118 and 14, 465 captions for training, respectively, whereas 413, 915 captions are provided in COCO caption [18] for image captioning. Due to the limited dataset, prior arts design decoders with shallow layers. Thus, the decoder may fail to learn generalized language expressivity but have a high chance of being fitted to the small-scaled target dataset [19].

To overcome these challenges, we propose a module-based approach that facilitates learning rich audio caption generation even from a small-scaled dataset and achieves generalization better than the prior arts. Rather than jointly training shallow decoder and encoder layers from scratch for caption generation, we leverage a powerful pre-trained language model (*e.g.*, GPT2 [20]), which is trained with a massive text-only dataset for text generation. What we only learn is to extract audio features containing global and temporal information of the audio and condition those to the frozen language model for caption generation. However, directly feeding these features produces poor caption qualities due to the modality gap between audio and text. To smoothly associate such gap, we employ mapping networks that translate audio features to the form that the language model can understand, called “prefix”.

The most closely related work is Koizumi *et al.* [21], which utilizes GPT2 as a caption generator. While sharing the same motivation, they still heavily depend on the target dataset by retrieving similar captions from the training dataset from the perspective of audio. All the retrieved captions from the training dataset are fed to GPT2 to guide text generation. On the contrary, our approach is inspired by recent success in prefix tuning [22, 23], which allows us to efficiently adapt an expressive language model to new text generation tasks by simply feeding learned prefixes. We conduct extensive experiments on Clotho and AudioCaps and show the effectiveness of our method compared to the prior arts. We further measure the text generation quality of our method from different perspectives, such as text-based audio retrieval and image generation from sound through text.

2. METHOD

2.1. Overall architecture

Our goal is to design a generalizable audio captioning model by leveraging the expressive pre-trained language model. The overall architecture is shown in Fig. 1-(a). Our model consists of an audio encoder that extracts features from the input audio, two mapping networks that translate the extracted features to the prefixes, and the language model which produces the texts from the prefixes.

Given a spectrogram converted from an input waveform, audio encoder $\text{Enc}(\cdot)$ extracts two types of features:

$$[f_t, f_g] = \text{Enc}(x), \quad (1)$$

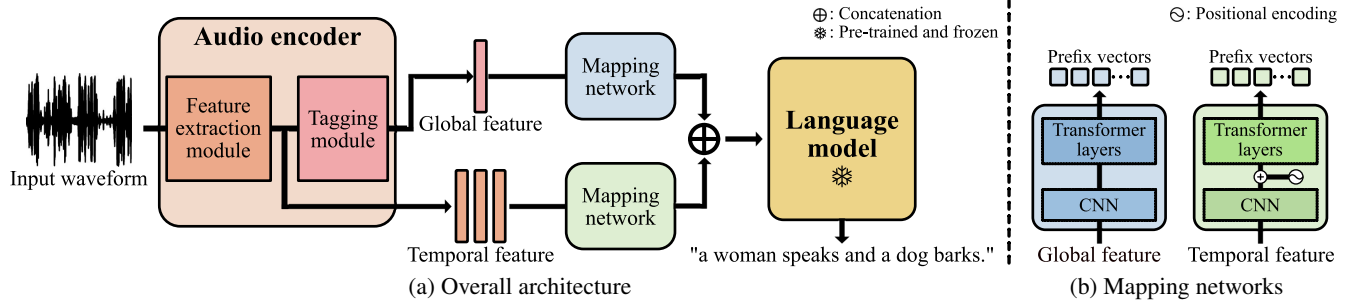
where x denotes the input spectrogram, and f_t and f_g are temporal and global audio features, respectively. Then, the features are mapped to the prefixes by using the mapping networks M . followed by concatenation:

$$[p_1^t, \dots, p_n^t] = M_t(f_t), \quad [p_1^g, \dots, p_m^g] = M_g(f_g), \quad (2)$$

$$V = [p_1^t, \dots, p_n^t, p_1^g, \dots, p_m^g], \quad (3)$$

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¹We tested on the other methods in which codes are publicly available.



where p^t and p^g are prefixes of length n and m , respectively, and have the same dimension as word embeddings. Finally, the concatenated prefixes V are fed to the auto-regressive language model and generate text descriptions corresponding to the input sound.

Among the modules, the audio encoder and the mapping networks are trained while fixing the language model to be unchanged. These trainable modules are trained to predict future word tokens y_i given the conditioning prefixes V and previously generated tokens. The language model is kept frozen during training, but gradients are back-propagated through the language model to the end; thereby, the audio encoder and the mapping networks are updated to learn prefix mappings understandable by the language model. In this way, the frozen language model can maintain its expressivity on the text generation task. Thus, our learning objective is to minimize the auto-regressive cross-entropy loss of the T length generated tokens:

$$L = - \sum_{i=1}^T \log p(y_i | y_{<i}, V). \quad (4)$$

2.2. Module details

Language model. We utilize a powerful auto-regressive language model, GPT2² [20], as our decoder, which is trained on a large-scale dataset (denoted as WebText) collected from the internet. Several studies [22, 23] have shown that this strong pre-trained language model can be adapted to new text generation tasks by simply feeding learnable prefixes. Motivated by this, we postulate that the audio features can be transformed into prefixes specific to the audio captioning task. Thus, we translate the audio features obtained from the audio encoder to prefixes through the mapping networks and feed the prefixes to the GPT2 for generating captions.

Audio encoder. We initialize the weights of the audio encoder by the pre-trained audio neural networks (PANNs) [5] with 14-layer CNN (CNN14), which is trained on AudioSet [24] to predict audio taggings. CNN14 contains a feature extraction module that extracts the feature map from the spectrogram and a tagging module that predicts audio tags from the extracted feature map. From a fixed-sized input spectrogram, CNN14 first extracts a $N \times 2 \times 2048$ sized feature map, where N , 2, and 2048 are time, frequency, and channel dimensions, respectively. We denote this feature map as *temporal feature*. This feature map is globally averaged and fed to the tagging module to produce a single vector, denoted as *global feature*. We assume that the temporal feature can provide fine-grained local information

along the temporal axis, while the global feature contains the global context information of the audio.

Mapping networks. We design the mapping networks to bridge the gap between audio features and GPT2. To handle the varying audio length, we concatenate a fixed length of learnable tokens along with each of the global and temporal features and feed them to the Transformer layers of respective mapping networks. From the output tokens of the mapping networks, we discard the tokens corresponding to the global and temporal features and only use the learnable tokens as prefixes for GPT2. Therefore, the number of prefixes remains the same no matter how long the length of input audio is. Each audio feature is translated to $K \times 768$ sized prefixes, where K is the number of prefixes depending on each mapping network, and 768 is the channel dimensions. Both mapping networks share the same CNN-Transformer design as shown in Fig. 1 (b), except that the mapping network for the temporal feature uses positional encoding to provide temporal position cues to the mapping network.

2.3. Implementation details

For training on the Clotho [17] dataset, the input of the audio encoder is a 1500×64 log mel spectrogram converted from 30 seconds of audio. We zero-pad the audio if the audio length is shorter than 30 seconds. We set the batch size to 55 and a weight decay to 0.02. For training on the AudioCaps [16] dataset, the input of the audio encoder is a 500×64 log mel spectrogram converted from 10 seconds of audio, and we set the batch size of 75 and a weight decay to 0.01. In all experiments, we use the AdamW optimizer, set the learning rate to peak on $5 \cdot 10^{-5}$ and gradually decay, and train our model for 50 epochs with early stopping.

3. EXPERIMENTS

3.1. Evaluation settings

Datasets. We conduct experiments using two benchmarks, AudioCaps [16] and Clotho [17]. AudioCaps is currently the largest audio captioning dataset, containing around 50K of 10 second in-the-wild audio clips sourced from Audioset [24]. Each audio is annotated with one caption in the training set and five captions in the evaluation set. Clotho consists of 4981 audio samples of 15 to 30 seconds duration. Each audio is annotated with five captions. We follow the standard protocols of training, validation, and test splits on each dataset for the evaluations.

Compared methods. We compare our proposed approach with six competing methods. Eren *et al.* [10] are the first to use PANN [5] for

²We use the pre-trained GPT2 from: <https://huggingface.co/gpt2>

Table 1. Evaluation on Clotho. Each method is trained on three different settings and tested on the Clotho dataset.

Training set	Method	BLEU ₁	BLEU ₂	BLEU ₃	BLEU ₄	METEOR	ROUGE _L	CIDEr	SPICE	SPIDEr
(i) Clotho	Mei <i>et al.</i> [14]	0.527	0.327	0.211	0.131	0.158	0.356	0.320	0.105	0.213
	Gontier <i>et al.</i> [9]	0.506	0.318	0.210	0.134	0.148	0.338	0.278	0.092	0.185
	†Chen <i>et al.</i> [11]	0.534	0.343	0.230	0.151	0.160	0.356	0.346	0.108	0.227
	†Xu <i>et al.</i> [12]	0.556	0.363	0.242	0.159	0.169	0.368	0.377	0.115	0.246
	†Koh <i>et al.</i> [13]	0.551	0.369	0.252	0.168	0.165	0.373	0.380	0.111	0.246
	Ours	0.560	0.376	0.253	0.160	0.170	0.378	0.392	0.118	0.255
(ii) AudioCaps	Mei <i>et al.</i> [14]	0.294	0.146	0.080	0.043	0.096	0.239	0.117	0.050	0.084
	Gontier <i>et al.</i> [9]	0.309	0.146	0.071	0.034	0.098	0.233	0.112	0.046	0.079
	Chen <i>et al.</i> [11]	0.226	0.114	0.065	0.039	0.086	0.228	0.109	0.042	0.076
	Ours	0.342	0.195	0.115	0.065	0.112	0.276	0.192	0.074	0.133
(iii) Clotho & AudioCaps	Mei <i>et al.</i> [14]	0.516	0.318	0.204	0.127	0.157	0.351	0.313	0.105	0.209
	Gontier <i>et al.</i> [9]	0.461	0.282	0.182	0.117	0.136	0.318	0.251	0.083	0.167
	Chen <i>et al.</i> [11]	0.516	0.325	0.215	0.141	0.153	0.350	0.314	0.102	0.208
	Ours	0.539	0.346	0.227	0.142	0.159	0.366	0.319	0.111	0.215

† are directly quoted from the original papers, or we trained the models using the publicly available codes otherwise.
We highlight the best results in **bold** and underline the second best among all the methods.

Table 2. Evaluation on AudioCaps. Each method is trained on three different settings and tested on the AudioCaps dataset.

Training set	Method	BLEU ₁	BLEU ₂	BLEU ₃	BLEU ₄	METEOR	ROUGE _L	CIDEr	SPICE	SPIDEr
(i) AudioCaps	†Mei <i>et al.</i> [14]	0.647	0.488	0.356	0.252	0.222	0.468	0.679	0.160	0.420
	†Gontier <i>et al.</i> [9]	0.699	0.523	0.380	0.266	0.241	0.493	0.753	0.176	0.465
	Chen <i>et al.</i> [11]	0.550	0.385	0.264	0.178	0.173	0.390	0.443	0.117	0.280
	†Eren <i>et al.</i> [10]	0.710	0.490	0.380	0.230	0.290	0.590	0.750	-	-
	Ours	0.713	0.552	0.421	0.309	0.240	0.503	0.733	0.177	0.455
(ii) Clotho	Mei <i>et al.</i> [14]	0.415	0.219	0.121	0.063	0.134	0.303	0.149	0.066	0.107
	Gontier <i>et al.</i> [9]	0.425	0.223	0.124	0.061	0.128	0.298	0.147	0.060	0.104
	Chen <i>et al.</i> [11]	0.365	0.170	0.091	0.048	0.110	0.273	0.083	0.049	0.066
	Ours	0.449	0.266	0.157	0.084	0.144	0.330	0.211	0.083	0.147
(iii) Clotho & AudioCaps	Mei <i>et al.</i> [14]	0.682	0.507	0.369	0.266	0.238	0.488	0.701	0.166	0.434
	Gontier <i>et al.</i> [9]	0.635	0.461	0.322	0.219	0.208	0.450	0.612	0.153	0.383
	Chen <i>et al.</i> [11]	0.489	0.292	0.178	0.106	0.152	0.346	0.265	0.093	0.179
	Ours	0.708	0.547	0.402	0.283	0.238	0.499	0.710	0.167	0.438

† are directly quoted from the original papers, or we trained the models using the publicly available codes otherwise.
We highlight the best results in **bold** and underline the second best among all the methods.

the audio feature extractor and feed the extracted features to the Bidirectional Gated Recurrent Unit (BiGRU) model. Koh *et al.* [13] and Chen *et al.* [11] extract audio features in a similar way, but train the Transformer decoder to generate captions, while Xu *et al.* [12] design a GRU for the decoder. Gontier *et al.* [9] leverage a pre-trained language model based on Bidirectional and Auto-Regressive Transformers (BART) [25], and finetune it for audio captioning tasks. Finally, Mei *et al.* [14] design the convolutional free transformer-based architecture to better capture global information as well as the temporal relationships between audio events.

Experiment setups and metrics. We conduct the experiments in three setups; (i) using training and test sets from a homogeneous dataset, *e.g.*, both training and test sets from AudioCaps, (ii) using ones from heterogeneous datasets, *e.g.*, the training set from AudioCaps and the test set from Clotho, and (iii) training on all the available datasets. All the prior arts focus on (i), while we propose that (ii) and (iii) would be more practical settings to evaluate the generalizability of the audio captioning models. For (ii) and (iii), we only evaluate the other methods in which the codes are publicly available. While our model generates plausible captions with the above settings, the discrepancy between the GPT2 and the target datasets' vocabulary sets may hinder the model from using expected wording by test samples. Therefore, we further train the GPT2 header to map the output embeddings to the vocabulary set of the target dataset only

in setting (i). All the methods are evaluated by the metrics widely used in the captioning tasks, including BLEU [26], METEOR [27], ROUGE-L [28], CIDEr [29], SPICE [30], and SPIDEr [31].

3.2. Comparisons

We compare our proposed model with the competing methods according to the settings described in Sec. 3.1. As shown in (i) of both Table 1 and Table 2, our model shows favorable performance against the other methods. In setting (ii), where the test set distribution is different from the training one, the other methods show significant performance drops in all the metrics, while our method favorably defends the drop. We postulate that the other methods may be fitted to the target dataset by training the decoder with the target dataset. Finally, we conduct an experiment (iii) to further evaluate whether the methods can be generalized to the overall datasets. As shown in both tables, our model outperforms all the metrics by a noticeable margin. Interestingly, our model in (iii) shows comparable results with that of (i), while others show a notable drop compared to their results in (i). This may hint that the frozen language model could prevent the model from overfitting to the target dataset in data-scarce settings.

In addition, we quantitatively evaluate the quality of the audio captions from another perspective by measuring text-based audio retrieval performance (Table 3). Rather than directly embed-

Table 3. Text-to-audio retrieval (R@k in %) results on Clotho. Note that Onescu *et al.* [3] is particularly designed for this task, while ours and the other methods target the audio captioning task.

Method	R@1	R@5	R@10
Mei <i>et al.</i> [14]	4.0	14.1	21.6
Gontier <i>et al.</i> [9]	2.1	7.0	12.0
Chen <i>et al.</i> [11]	1.5	4.4	7.5
Ours	7.6	19.6	28.8
†Onescu <i>et al.</i> [3]	4.0	-	25.4

† is directly quoted from the original paper.

Table 4. Ablation studies of our proposed method. We compare the different configurations of our method on AudioCaps. M denotes the mapping network, and f_t and f_g denote temporal and global feature, respectively.

Design choice			Metrics				
M	f_t	f_g	BLEU ₄	METEOR	ROUGE _L	CIDEr	SPIDEr
	✓	✓	0.250	0.216	0.471	0.614	0.387
✓	✓		0.280	0.230	0.491	0.708	0.440
✓		✓	0.297	0.230	0.485	0.652	0.413
✓	✓	✓	0.309	0.240	0.503	0.733	0.455

ding text and audio into a joint embedding space, we apply our method to audio in a database (DB) so that we can construct an audio-text pair DB. Thereby, the generated texts can substitute audio in text-to-audio, which results in text-to-text retrieval. We use MP-Net [32] for measuring the text similarities on text-to-text retrieval. Our method provides promising results compared to the other methods when evaluated on the test text queries. In particular, our method performs favorably against the baseline model of Onescu *et al.* [3], which is specifically designed for the text-to-audio retrieval task.

3.3. Ablation studies

We conduct ablation studies to evaluate our design choices; the effectiveness of the mapping network, and the synergy of using both global and temporal features. The results are summarized in Table 4. We observe that using both global and temporal features may further enhance performance. However, using both features without mapping networks causes a significant performance drop. This result shows that the mapping network acts as a crucial role in bridging the gap between two heterogeneous modalities, *i.e.*, audio and text.

3.4. Qualitative results

As shown in Fig. 2, our model produces plausible captions on all the experimental settings in 3.1. Even though tested on the unseen dataset as in (ii), our model captures salient events in the audio and expresses them as text descriptions. Furthermore, the generated texts are expressive enough to be extended to visualize the audio signals through the recent text-to-image model [33].

4. CONCLUSION

In this paper, we present a compositional module-based approach to achieve a generalizable audio captioning model given a small-scaled dataset. We observe that most recent audio captioning methods train whole models, including the caption generator parts (decoder), and tend to be overfitted to the target dataset due to the lack of an audio-text paired dataset. To overcome such a problem, we leverage a pre-trained language model as our decoder and keep it frozen to maintain its expressivity for the text generation task. Instead, we train

	(a) kXjzsroVTtw.wav	(b) Lluvia 1.wav
GT	a man speaks with small birds chirping in the distance	a rumbling sound of thunder with rain falling heavily in the background
(i)	a man speaks as birds chirp in the background	a heavy rain is falling down and thunder rumbles
(ii)	birds are chirping and a man is speaking in the background	rain falls hard and thunder roars in the distance
(iii)	a man speaking followed by birds chirping	rain is falling and thunder is rumbling in the background

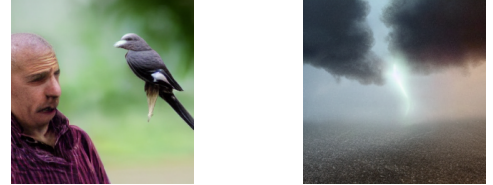


Fig. 2. Sample results of audios (a) and (b). [Top] The results show that our model can produce plausible captions even on the out-of-the-domain dataset in (ii). The terms (i), (ii) and (iii) refer to the same settings in Tables 1 and 2. [Bottom] We extend our method to visualize the sound scene by feeding the generated texts of (i) to the text-to-image model [33]. The audios (a) and (b) are sampled from the test sets of the AudioCaps and Clotho datasets, respectively.

our audio encoder to extract global and temporal features from input audio and train the mapping networks to translate these features to the prefixes. These prefixes are then fed to the frozen language model and generate a caption corresponding to the input audio. We further demonstrate that our method can be extended to diverse applications, such as text-based audio retrieval and image generation from sound through text. For future work, compensating for the lack of audio-text pairs by visual modality [34], dense prediction of captioning [35, 36], or leveraging large-scaled pseudo dataset [37, 38] for the audio captioning task would be interesting to investigate.

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